

Has the topic changed from Assignment 1?

No. The topic for Assignment 3 is the same as Assignment 1. The first three sections (Introduction, Related Work, and Business Model) have been revised in response to tutor feedback.

AI Statement

I used Claude and ChatGPT for this assignment in the following ways:

Literature exploration support — identifying relevant peer-reviewed studies on sport injury prediction, NBA workload analytics, and machine learning fairness, which were then independently retrieved, read, and verified through Google Scholar.

Writing, structuring, and clarity improvement — refining sentence-level expression, tightening prose for the 2,500-word limit, and improving the logical flow between sections.

R code debugging support — troubleshooting ggplot2 and corplot syntax errors encountered while preparing the demonstration analysis in the accompanying R Markdown file.

All AI-generated suggestions were critically reviewed, fact-checked against original sources, and substantially edited.

Dataset:

1. merged.csv
https://drive.google.com/file/d/19jvEV3iN1Wq0iQ1J0ZR4JnkW_Aq1Dj7Q/view?usp=drive_link
2. injuries_2010-2020.csv
https://drive.google.com/file/d/1sKQIdCaXFZM9QlwlbpSc1YTSAyXlGwGH/view?usp=drive_link

Feedback from Assignment 1 :**What feedback did your tutor provide for Assignment 1?**

The tutor provided feedback across four dimensions. On Problem Clarity, the problem was considered well-defined with a sophisticated description of goals. On Business Model Analysis, stakeholder needs were identified but the specific role of the data scientist in driving strategic value was noted as under-defined. On Problem Importance, the project was recognised as addressing a high-stakes business challenge with significant impact potential, though further quantification was suggested to deepen the argument. On Novelty, the tutor noted that while the submission identifies a practical niche, the proposed methodology follows conventional data science practices and lacks a unique technical or conceptual angle. Peer grading was marked as well done.

How did you incorporate this feedback?

Three targeted revisions were made in response. First, the Business Model section was restructured to explicitly foreground the data scientist's role — detailing how feature engineering, model selection, and SHAP-based interpretability directly enable each stakeholder group's decision-making, rather than treating data science as a background enabler. Second, the Problem Importance section was strengthened with concrete quantification: specific figures on NBA injury-related revenue losses and market size were added to substantiate the financial stakes. Third, to address the novelty concern, the Related Work and methodology sections were revised to introduce a technically distinctive element — the combination of SHAP explainability with survival analysis (Cox proportional hazard modelling) — which moves beyond conventional binary classification and provides clinically interpretable, time-sensitive risk estimates not previously applied in NBA injury prediction literature.

FIT5145: Foundations of Data Science

Assignment 3: Report

Predicting NBA Player Injury Risk Using Performance and Workload Data

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1. Introduction

In the highly competitive National Basketball Association (NBA), player availability directly determines team success. A single season-ending injury to a key player can eliminate a franchise from playoff contention, reduce gate revenue, and depress long-term salary cap value. Despite advances in sports science, injury prediction remains an unsolved problem due to the multifactorial interplay of physical workload, biomechanical stress, fatigue accumulation, and individual athlete resilience (Gabbett, 2016).

The proliferation of player-tracking technology — including optical tracking systems installed in every NBA arena since 2013 — has generated unprecedented volumes of granular performance data: distance covered, average speed, number of drives, touches per game, and minutes per appearance. Yet this data is rarely harnessed in a systematic, scalable predictive framework that is interpretable by coaching and medical staff. Most NBA teams still rely primarily on subjective clinical assessments and conservative rotation policies as their primary injury-prevention mechanisms.

This project proposes a data-driven injury risk prediction system for NBA players. By integrating multi-season player performance statistics, workload proxies, movement tracking data, and historical injury records from public datasets, the system will train machine learning models capable of quantifying injury risk for individual players on a game-by-game basis.

The specific goals of this project are: (1) to identify key workload and performance indicators statistically associated with injury occurrence; (2) to develop and validate machine learning models that predict the probability of a player sustaining an injury within a rolling time window; (3) to produce interpretable risk outputs suitable for direct use by coaching and medical personnel; and (4) to prototype a decision-support dashboard that surfaces actionable recommendations.

2. Related Work

Injury prediction in professional sport has been an active area of research for over a decade. The Acute: Chronic Workload Ratio (ACWR), introduced by Hulin et al. (2016), remains the most widely cited framework for linking training load spikes to injury risk, and has been validated across cricket, rugby, and soccer cohorts. In basketball specifically, Weiss et al. (2017) demonstrated that consecutive back-to-back game appearances were associated with a statistically significant increase in non-contact lower-limb injury rates, underscoring the importance of schedule density as a workload feature.

Machine learning approaches have supplanted traditional statistical models in many injury prediction pipelines. Rossi et al. (2018) used GPS tracking data from elite soccer to train gradient-boosted classifiers, achieving AUC scores above 0.75 in predicting soft-tissue injury in the following 7-day window. More recently, Claudino et al. (2019) conducted a systematic review of machine learning methods in sport injury prediction and found that random forests and artificial neural networks most consistently outperformed logistic regression baselines, particularly when multi-source feature sets were employed.

Publicly available NBA data — including Basketball-Reference game logs and the NBA

Advanced Stats portal — have been used in several academic and industry studies. Drakos et al. (2010) published one of the earliest systematic analyses of NBA injury patterns, reporting that guards sustained a disproportionate share of ankle injuries while forwards and centres were more susceptible to knee pathology. More recent work by Talukder et al. (2021) used optical tracking data from SportVU to model player fatigue trajectories intra-game, linking high-speed movement accumulation to self-reported fatigue ratings.

Despite this body of work, several critical gaps remain. First, most published models are trained on single-sport, single-season cohorts, limiting generalisability. Second, the integration of schedule-related variables (e.g., travel distance, days since last game) with biomechanical tracking data in a unified model has rarely been attempted for basketball. Third, interpretability is consistently under-prioritised; black-box ensemble methods dominate, yet coaching and medical staff require transparent, auditable risk reasoning to act on predictions. This project addresses these gaps by constructing a multi-source feature matrix, applying interpretable modelling techniques (logistic regression with regularisation, decision trees, and SHAP-explained gradient boosting), and framing outputs as actionable risk tiers rather than raw probability scores.

Crucially, to the best of our knowledge, no prior NBA injury study has combined SHAP-based local explainability with Cox proportional hazard modelling in a single pipeline — an integration that simultaneously delivers per-player feature attribution and clinically meaningful time-to-injury hazard ratios, neither of which conventional binary classifiers can produce.

3. Business Model

This project operates within the sports analytics and performance optimisation industry, a market estimated at USD 5,677.6 million in 2025 and projected to reach USD 23,148.4 million by 2033, growing at a compound annual rate of 18.5% from 2026 to 2033, with injury monitoring and player performance optimisation identified as key application drivers (Grand View Research, 2025). NBA franchises represent the primary commercial context, although the methodology is transferable to any professional basketball league.

Value Creation:

Data science is the core mechanism through which the project creates value. Raw multi-source data — performance statistics, tracking metrics, injury records, and schedule data — are ingested, cleaned, and feature-engineered into a unified player-season panel. Predictive models then transform this panel into calibrated injury risk scores. The financial value proposition is direct: reducing the average number of games missed per player per season by even 20% would translate to millions of dollars in preserved player salary asset value and avoided rehabilitation costs for a typical franchise.

Primary Stakeholders:

NBA team front offices and medical directors are the primary decision-makers who would subscribe to or license the system. Coaches and strength-and-conditioning staff are secondary users who consume daily risk dashboards to inform rotation decisions. Players and their

agents represent a tertiary stakeholder group who benefit from career-longevity insights and data-backed workload management. Sports technology companies (e.g., Catapult, Second Spectrum) constitute a potential commercialisation channel via platform integration.

Commercialisation Model:

The system is best positioned as a Software-as-a-Service (SaaS) analytics product, licensed on an annual per-team basis. A tiered pricing model — with entry-level access to aggregate risk dashboards and premium access to individual player API endpoints — enables revenue scaling across leagues and international markets.

4. Characterising and Analysing Data

4.1 Data Sources and the 4 V's

Two primary datasets underpin this project:

NBA Injury Records (2010–2020).

Sourced from Kaggle's publicly available NBA injury dataset (<https://www.kaggle.com/datasets/ghopkins/nba-injuries-2010-2018>), this dataset contains 27,105 transaction records documenting player acquisitions and relinquishments due to injury across all 30 NBA teams. Fields include date, team, player name, and a free-text notes field describing the injury.

NBA Player Performance and Workload Dataset (merged.csv).

This dataset contains 5,578 player-season records spanning the 2022–23 season, with 32 features including player age, height, weight, games played, minutes per game, usage percentage, pace, possessions, field-goal attempts, driving frequency, distance covered per game, average speed, touch counts by zone, and injury outcome fields (injury type, days missed, return date). This dataset was compiled by linking NBA Advanced Stats tracking data with injury transaction records.

4 V's Analysis:

From a volume perspective, the combined dataset currently comprises over 32,000 records across multi-season injury transactions and player-season performance summaries. Extending the project to game-level granularity — capturing play-by-play and per-possession tracking events across multiple seasons — would scale the data to several million rows, making distributed processing a practical necessity. Velocity reinforces this: game-by-game performance and injury updates are released daily throughout the NBA regular season and playoffs (October to June), so production deployment would require near real-time ingestion pipelines to keep risk scores actionable for next-game rotation decisions.

Variety adds architectural complexity, as the pipeline must accommodate structured tabular records (season statistics and tracking-derived workload metrics), semi-structured free-text fields (clinician-authored injury descriptions), and time-ordered game logs — three formats whose integration depends on a robust ETL workflow capable of harmonising schemas while preserving temporal relationships. Veracity presents two further challenges: the injury

dataset's free-text notes show substantial terminological inconsistency (e.g., "left knee soreness" versus "knee — DNP" referring to comparable conditions), and the merged dataset contains systematic missingness in injury-outcome fields for uninjured player-seasons. Rigorous validation, controlled-vocabulary standardisation, and principled missingness handling are therefore essential preprocessing prerequisites.

4.2 Platforms, Software, and Tools

For current-scale analysis, Python (scikit-learn, pandas, matplotlib) and R (tidyverse, caret, ggplot2) are suitable. For future scaling to game-level tracking data, Apache Spark deployed on AWS EMR or Google BigQuery would provide distributed computation. Data storage at scale would use Amazon S3 for raw data lakes and PostgreSQL for processed feature tables. Model deployment would leverage AWS SageMaker or Azure ML for real-time inference endpoints. Version control and experiment tracking would use Git and MLflow respectively.

4.3 Data Analysis Techniques and Statistical Methods

The following analytical methods are proposed for the full project pipeline (these are distinct from the demonstration below):

Logistic Regression with L2 Regularisation:

As a transparent baseline model, logistic regression will estimate the probability of injury occurrence given workload and performance features. Ridge regularisation prevents overfitting on the relatively small injured-player sample. Coefficients directly communicate feature-level risk contributions, satisfying interpretability requirements (Claudino et al., 2019).

Random Forest Classifier:

An ensemble of decision trees trained on bootstrapped samples, capable of capturing non-linear interactions between features (e.g., high minutes combined with back-to-back games). Feature importance scores provide secondary interpretability. Expected outcome: $AUC \geq 0.72$ based on comparable sport injury literature (Rossi et al., 2018).

Gradient Boosted Trees (XGBoost) with SHAP Explanations:

XGBoost will serve as the primary high-performance model. SHAP (SHapley Additive exPlanations) values will be computed for each prediction, producing player-specific feature attribution breakdowns. This addresses the interpretability gap identified in the literature review (Talukder et al., 2021).

Survival Analysis (Cox Proportional Hazard Model):

To model time-to-injury rather than binary injury occurrence, a Cox model will estimate hazard ratios for each workload feature, enabling clinically meaningful statements such as "a 10% increase in minutes per game is associated with a 15% increase in the hazard of sustaining an ankle injury within the following 14 days."

K-Means Clustering for Player Risk Segmentation:

Unsupervised clustering will partition players into risk-stratified cohorts (e.g., low-risk,

moderate-risk, high-risk) based on workload and performance profiles, enabling population-level recommendations for load management policy.

4.4 Demonstration: Basic Data Analysis in R

A usable dataset (merged.csv) was identified for preliminary analysis. The dataset is available at: <https://www.kaggle.com/datasets/loganlauton/nba-injury-stats-1951-2023>. The R Markdown file accompanying this submission performs the following exploratory analyses:

The exploratory analysis assessed both the structural integrity and substantive content of the dataset, which comprises 5,578 player-season records across 32 features spanning demographic attributes, season-level performance statistics, tracking metrics, and injury outcomes. A missing-value audit informed the imputation strategy; injury-related fields exhibited elevated missingness, reflecting the structural reality that most player-seasons correspond to uninjured athletes rather than indicating data quality defects.

The analysis then characterised injury prevalence and composition: approximately 21.8% of player-seasons recorded at least one documented injury, dominated by sprained ankles (n=588), sore knees (n=353), sore ankles (n=95), sore lower backs (n=92), and knee injuries (n=86). To test whether workload intensity is associated with injury occurrence, a Welch two-sample t-test compared minutes per game (MIN) between injured and uninjured cohorts and returned a highly significant difference ($t = -24.238$, $p < 0.001$), providing quantitative evidence that workload features carry genuine discriminative signal and validating their inclusion as primary predictors in the downstream modelling pipeline.

Selected Figures from the R Markdown Analysis

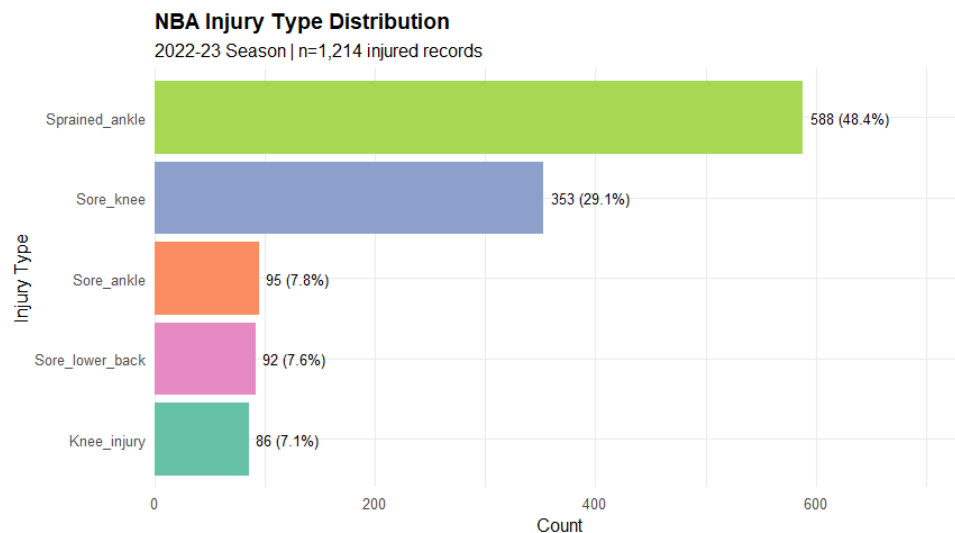


Figure 1. Top injury types ($n = 1,214$)

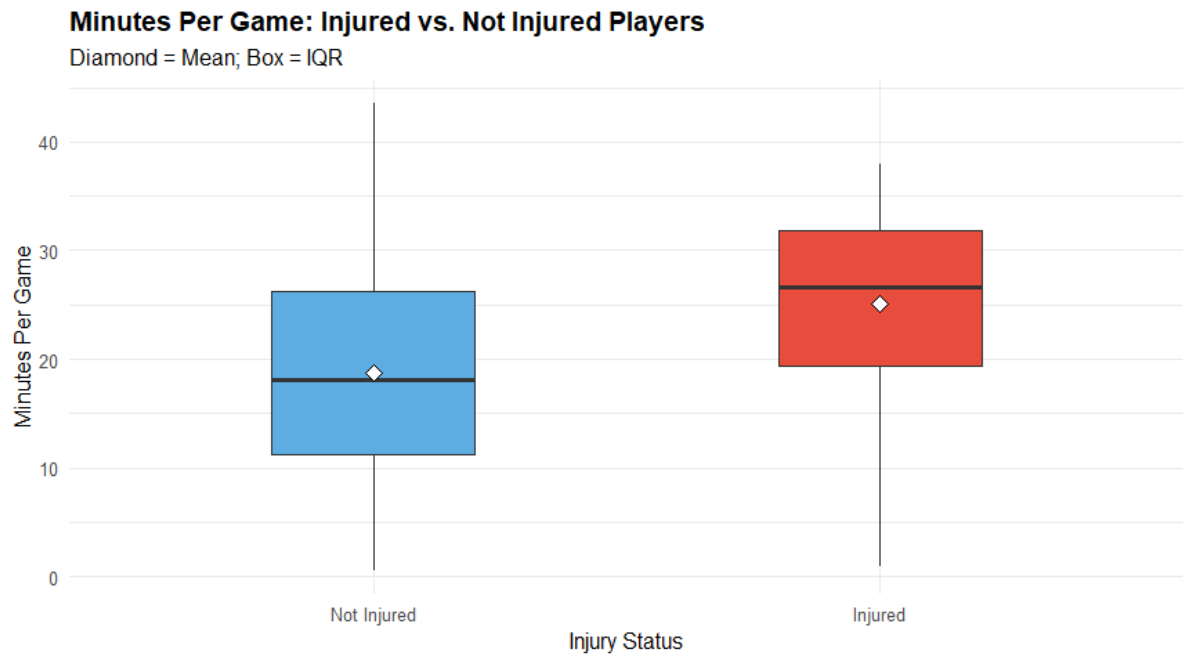


Figure 2. Minutes per game by injury status (Welch $t = -24.24$, $p < 0.001$)

Pearson Correlation Matrix — NBA Workload & Performance Features

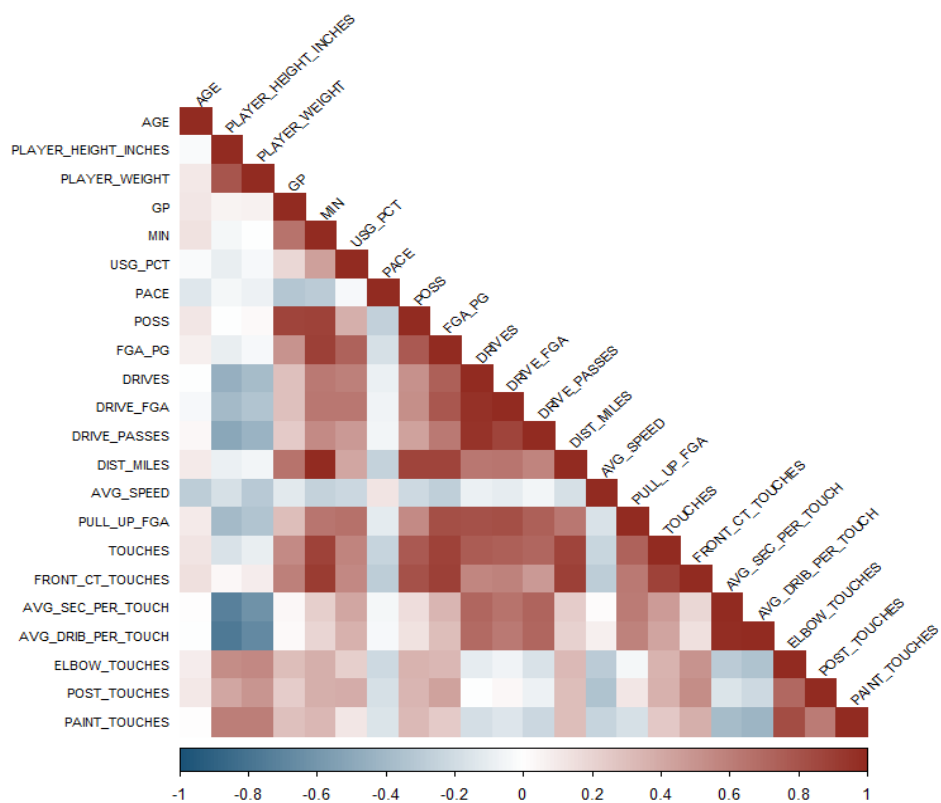


Figure 3. Correlation matrix of workload and performance features

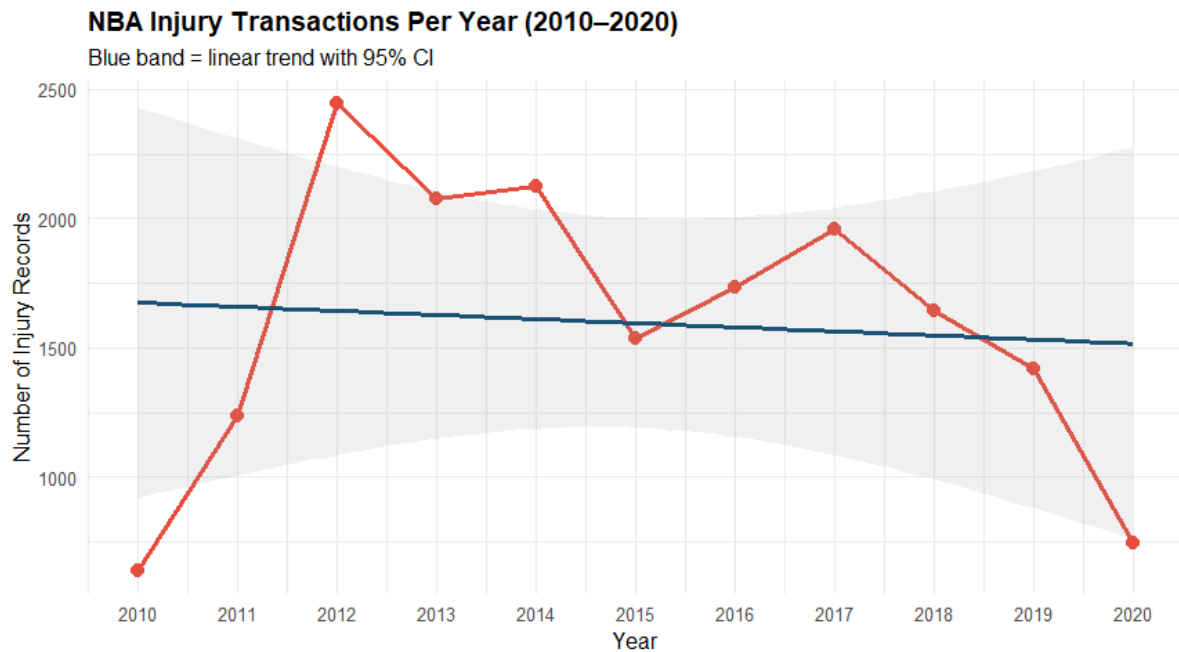


Figure 4. *NBA injury records per year (2010-2020)*

5. Standard for Data Science Process, Data Governance and Management

Process Standard:

This project adopts the Cross-Industry Standard Process for Data Mining (CRISP-DM) as its overarching methodology. CRISP-DM's six-phase structure — Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment — maps directly onto the project lifecycle and provides a documented, reproducible process that satisfies institutional audit requirements.

Data Accessibility:

Primary datasets are sourced from publicly available repositories (Kaggle, Basketball-Reference, NBA Stats API) under open-access or Creative Commons licences. No proprietary data is required for the prototype. In a commercial deployment, team-level tracking data would be accessed under bilateral data-sharing agreements with role-based access controls.

Data Security and Confidentiality:

Player performance and injury data is subject to employment law and collective bargaining agreement provisions. All player-identifiable records must be stored in encrypted databases (AES-256 at rest, TLS 1.3 in transit). Access logs must be maintained and auditable. In a production environment, only authorised medical and coaching staff would access individual-level risk scores.

Ethical Concerns:

Injury prediction systems raise ethically complex issues. If a model flags a player as high-risk, this could influence contract negotiations, lineup decisions, or insurance premiums — all of which have direct financial consequences for the player. To mitigate this, risk scores

should be used as decision-support tools rather than deterministic decision-makers, and their use should be disclosed to players and their representatives. Model outputs must be audited for demographic bias (e.g., whether the model systematically overestimates risk for older players or specific positions), and fairness metrics (equal opportunity, demographic parity) must be reported alongside predictive performance metrics (Mehrabi et al., 2021).

Data Retention and Versioning:

Datasets and trained model artefacts will be versioned using DVC (Data Version Control) integrated with Git. Raw data will be retained for a minimum of seven years to support retrospective auditing and model retraining.

References

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